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Math 215A

Homework 6
Due: November 4, 2026

Rings are understood to be commutative, unless stated otherwise.

Problem 1

Let R be a noetherian ring. We have shown (or will show) that $X = \text{Spec}(R)$ can be written as the union of finitely many irreducible closed subsets, $X = X_1 \cup X_2 \cup \cdots \cup X_m$, such that X_i is not contained in X_j for any $i \neq j$. Show that such a decomposition of X is unique up to reordering the X_i's. It is called the *decomposition of X into its irreducible components.*

Solution. Assume the contrary, $A_1 \cup ... \cup A_a$ and $B_1 \cup ... \cup B_b$ are two decompositions of X into irreducible. Note the following that

$$A_i = \bigcup_{j=1}^{b} A_i \cap B_j$$

and that $A_i \cap B_j$ is closed. Assume the contrary that $S = \{j \in [b] : A_i \cap B_j \neq \emptyset\}$ doesn't have only 1 element. $|S| \neq 0 \Rightarrow A_i = \emptyset$ which is not true. So $|S| > 1$. Thus

$$A_i = \bigcup_{j \in S} A_i \cap B_j,$$

but this contradicts the assumption that A_i is irreducible. Thus by contradiction, $\exists j_i \in [b]$ such that $A_i \subseteq B_{j_i}$. Let $\varphi : [a] \to [b]$ where $i \to j_i$.

By a similar argument, there is a $\psi : [b] \to [a]$ where $j \to i_j$ such that $B_j \subseteq A_{j_i}$.

Assume the contrary, that $\psi(\varphi(i)) = k \neq i$. Thus $A_i \subseteq B_{\varphi(i)} \subseteq A_k$. This contradicts the maximality of A_i, meaning $\psi \circ \varphi = 1_{[a]}$. By a symmetric argument, $\varphi \circ \psi = 1_{[b]}$. Moreover this tells us that

$$A_i \subseteq B_{\varphi(i)} \subseteq A_i \subseteq B_{\varphi(i)} \Rightarrow A_i = B_{\varphi(i)}.$$

meaning φ is just a permutation.

Problem 2

Write $\mathbb{A}^3_{\mathbb{C}}$ for affine 3-space over $\mathbb{C}$, meaning Spec $\mathbb{C}[x, y, z]$. Let X be the closed subset $\{x^2 = yz, xz = z\} \subset \mathbb{A}^3_{\mathbb{C}}$. (As always, this means the subset where both equations hold.) Decompose X into its irreducible components.

Solution.

$$\begin{aligned} X &= \{x^2 - yz = 0, xz - z = 0\} \\ &= \{x^2 - yz = 0, z(x-1) = 0\} \\ &= \{x^2 - yz = 0, z = 0\} \cup \{x^2 - yz = 0, x - 1 = 0\} \\ &= \{x^2 = 0, z = 0\} \cup \{yz - 1 = 0, x - 1 = 0\} \\ &= \{x^2 = 0, z = 0\} \cup \{yz = 1, x = 1\} \\ &= \{x = 0, z = 0\} \cup \{yz = 1, x = 1\} \end{aligned}$$

Consider

$$\begin{gathered} \mathbb{C}[x, y, z] \to \mathbb{C}[y] \\ 1 \to 1 \quad x \to 0 \quad y \to y \quad z \to z, \end{gathered}$$

the kernel of such a map is (x, z), which therefore is prime. $\overline{\{(x, z)\}} = \{x = 0, z = 0\}$.

Consider

$$\begin{gathered} \mathbb{C}[x, y, z] \to \mathbb{C}[z] \\ 1 \to 1 \quad x \to 1 \end{gathered}$$

Problem 3

Let k be a field. Let R be the k-subalgebra of $k[x, y]$ generated by $y, xy, x^2y, \ldots$. Show that R is not finitely generated as a k-algebra. Is R noetherian? Also, describe in geometric terms (thinking of polynomials as functions) which polynomials are in the subring R.

Solution.